LEARNER GUIDE

Inequalities as regions on a graph

Inequalities · Lesson 4. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

On a graph, a rule such as $x > 2$ is true for a whole **region** of points — every point whose x-coordinate is greater than 2, at any height. The region is bounded by a straight line, here $x = 2$. The line is drawn **dashed** when points on it are not included (< or >) and **solid** when they are (≤ or ≥) — just like open and filled circles on a number line. Two or three rules together allow only the points in the **overlap** of their regions.

What you are learning to notice

- A rule about x alone gives a region beside a vertical line; a rule about y alone gives a region above or below a horizontal line.
- Dashed line: points on it are not included. Solid line: they are.
- To choose the side, test one point that is not on the line, such as (0, 0).
- A point is in a combined region only if it obeys **every** rule.

Moving through it, one step at a time

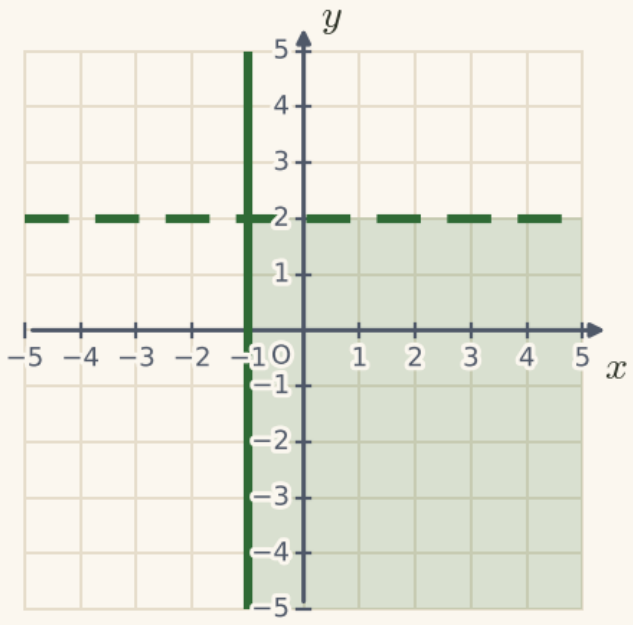
For each rule, draw its boundary line: replace the symbol with = (so $x + y < 5$ gives the line $x + y = 5$). Make it dashed or solid. Then test a point to decide which side is allowed. With several rules, the region is where all the allowed sides overlap. To list whole-number points, check each point carefully, especially any that lie on a boundary.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A region from two rules

Show the region where $x \geq -1$ and $y < 2$.

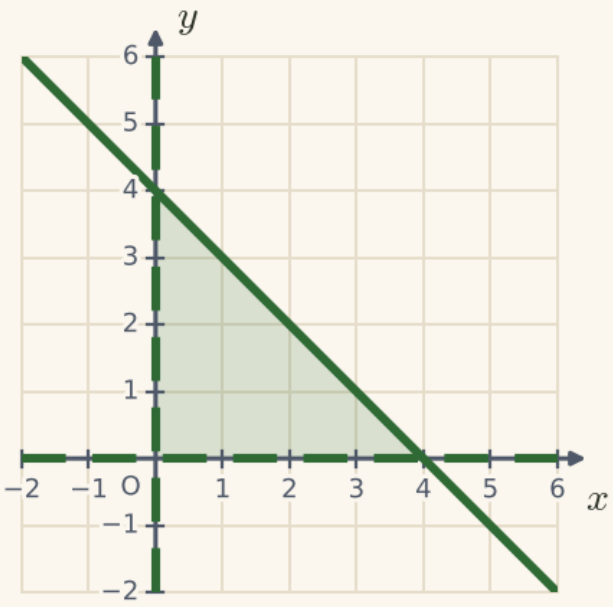


1. $x \geq -1$: draw the vertical line $x = -1$, solid (≥ includes it).	region to the right
2. $y < 2$: draw the horizontal line $y = 2$, dashed (< leaves it out).	region below
3. Shade where both are true.	the overlap

The region right of the solid line $x = -1$ and below the dashed line $y = 2$.

Example 2 — Counting whole-number points

How many whole-number points satisfy $x > 0$, $y > 0$ and $x + y \leq 4$?



1. $x > 0$ and $y > 0$: both coordinates at least 1.	axes left out
2. $x + y \leq 4$: the total is at most 4, so points on $x + y = 4$ count.	solid line
3. List them row by row.	(1, 1) (2, 1) (3, 1) (1, 2) (2, 2) (1, 3)

6 whole-number points.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (plotting coordinates in all four quadrants, and open and filled circles from Lesson 3), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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